The AI Negotiation Coach: Using Generative AI to Specifically Enhance Strategic Preparation, Strategy and Value-Creation in Business Negotiation Education

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***Abstract*—This study evaluates the efficacy of a generative AI negotiation coach through a true experimental design with Chinese business students. Results demonstrate that the AI group achieved significantly greater improvements in strategic preparation, strategy, and value creation compared to traditional methods, while the control group showed stronger development in communication skills. The AI coach effectively addressed critical gaps in strategic competency development, though human interaction remained essential for communication training. These findings support the targeted integration of AI in negotiation education, highlighting its value in cultivating strategic thinking and deal-design capabilities while underscoring the complementary role of traditional methods for interpersonal skill development.**

Keywords—generative ai, business negotiation education, exper imental study, strategic competencies, value creation

I. Introduction

The integration of artificial intelligence into higher education has gained significant momentum, driven by national policies advocating for technological innovation in cultivating talent. In China, initiatives such as the New Generation Artificial Intelligence Development Plan [1] and the 14th Five-Year Plan [2] explicitly emphasize the role of AI in transforming educational paradigms, fostering personalized learning, and enhancing students' practical competencies. Within business education, this vision aligns with efforts to modernize pedagogical approaches, particularly in domains like negotiation training, where strategic thinking and adaptive decision-making are critical for professional success [3].

Concurrently, generative AI (GenAI) has emerged as a transformative tool with the potential to create highly interactive and responsive learning environments [4]. Its ability to simulate complex, dynamic scenarios offers unprecedented opportunities for experiential learning—allowing students to engage in realistic, repeatable practice without the constraints of traditional classroom settings [5]. These advancements are especially relevant in negotiation education, where GenAI can emulate multifaceted negotiation counterparts, provide real-time feedback, and support the development of higher-order skills such as value creation, strategic planning, and communicative adaptability [6].

A persistent challenge in business negotiation education lies in the effective development of complex strategic competencies, particularly in the areas of thorough preparation and value-creating integrative negotiation. Traditional pedagogical methods, such as theoretical lectures and sporadic peer-based role-plays, often fall short; students lack sufficient opportunities for deliberate, repetitive practice in a risk-free environment, and they are frequently deprived of facing a sophisticated, adaptable opponent that can challenge their assumptions and expose strategic blind spots. The inadequacy of these methods is substantiated by compelling evidence from the Chinese context. According to the 2023 National Report on the Quality of Business Administration Education by the Ministry of Education, practical application abilities (including negotiation skills) remain the weakest link in talent development, with over 65% of surveyed enterprises expressing that recent graduates significantly lack the ability to handle complex business negotiations upon entering the workforce [3].

Furthermore, a research revealed that despite high theoretical scores, nearly 70% of graduates felt that classroom role-playing was vastly different from real commercial negotiations, citing a lack of "truly challenging opponents" and "high-stakes simulation environments" as the primary reasons [7]. This gap is further quantified by research published in China Higher Education, which found that in traditional teaching modes, the average student receives less than 15 minutes of personalized feedback throughout an entire semester of negotiation practice, severely limiting skill iteration and refinement [7]. These findings collectively underscore a systematic and severe disconnect within the current domestic educational paradigm, where an over-reliance on theoretical indoctrination and low-fidelity practice fails to cultivate the strategic agility required for real-world commercial success.

Recent scholarly work has begun to explore the potential of artificial intelligence, particularly generative AI, to address

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long-standing pedagogical limitations in experiential learning domains. Studies by [4] demonstrate that AI-driven simulations can significantly enhance learner engagement and skill acquisition in complex, interactive tasks such as negotiation. However, as [5] note, most existing applications remain narrowly focused on linguistic or procedural competencies rather than cultivating strategic depth or adaptive decision-making. In the specific context of negotiation training, [7] observed that while AI-based role-playing systems improve conversational fluency, they often lack the sophistication to simulate multi-issue, interest-based negotiations that require creativity and compromise. This sentiment is echoed in broader educational AI research, which calls for systems that emulate not only opponents but also coaches—capable of providing real-time feedback, provoking reflection, and strategically challenging learners' assumptions [8]. Yet, as [8] highlight in their review of AI in business education, few empirical studies measure the holistic impact of such systems across the full spectrum of negotiation competencies, particularly within non-Western educational contexts.

Based on the compelling evidence of systemic deficiencies in traditional negotiation pedagogy, a clear research gap emerges. While Generative AI (GenAI) has demonstrated significant potential in simulating interactive learning environments across various educational domains, existing applications in negotiation training have typically taken a narrow approach, focusing on isolated skills such as language expression or simple dialogue simulation, rather than offering a holistic framework that concurrently develops the interconnected competencies of preparation, strategy formulation, communication, adaptability, value creation, and outcome optimization essential for expert performance. Existing studies on AI in education often focus on general language practice or basic conversational agents, but there is a conspicuous lack of rigorous, empirical research on the use of dynamically adaptive AI opponents specifically designed to act as "coaches" to challenge and refine a learner's strategic reasoning and deal-design capabilities. Furthermore, despite the pronounced need within the Chinese educational context, no known studies have yet designed, implemented, and quantitatively evaluated a GenAI-driven pedagogical intervention aimed explicitly at bridging this well-documented gap between theoretical knowledge and practical strategic execution in negotiation. This study seeks to fill this void by investigating whether a purposely built AI negotiation coach can serve as a targeted solution to enhance these precise skills.

The primary purpose of this study is to design, implement, and rigorously evaluate a generative AI-powered negotiation coach, and to empirically investigate its efficacy as a pedagogical tool for developing a comprehensive set of negotiation competencies among business students in China. This study moves beyond generic AI chatbots by creating a system specifically engineered to provide deliberate practice across the interconnected domains of preparation, strategy, communication, adaptability, value creation, and outcomes.

RQ1: Are there any significant differences in post-test performance between students who trained with the generative AI negotiation coach and those who used traditional methods, as measured by their competencies in outcome (preparation, strategy, communication, adaptability, value creation)?

RQ2: Are there any significant differences between pre-test and post-test scores for students in the experimental group (AI coach training) across the competencies in outcome (preparation, strategy, communication, adaptability, value creation)?

RQ3: Are there any significant differences between pre-test and post-test scores for students in the control group (traditional role-playing) across the competencies in outcome (preparation, strategy, communication, adaptability, value creation)?

II. Methodology

This study employed a true experimental pretest-posttest control group design to evaluate the efficacy of a generative AI negotiation coach integrated into the Chaoxing learning platform. The study population consisted of undergraduate management students enrolled in a business negotiation course at a large university in Hunan Province, China. Participants were randomly assigned to either an experimental group ($n = 45$) or a control group ($n = 45$) through the university's registration system, ensuring equivalence between groups before the intervention. The experimental group utilized the AI negotiation coach during practice sessions, while the control group engaged in traditional peer-to-peer role-playing activities. Both groups completed identical pre-test and post-test assessments evaluated by three independent experts blinded to group assignment.

The pre-test and post-test employed the same negotiation case to ensure measurement consistency and avoid content-related bias. The selected case involved a multi-issue business negotiation scenario between a manufacturing supplier and a retail distributor, featuring six negotiable issues with varying priorities for each party: unit price, minimum order quantity, payment terms, delivery schedule, quality assurance specifications, and contract duration. This design created clear trade-off opportunities for value creation while maintaining sufficient complexity to assess all five competency dimensions.

The assessment procedure followed a standardized protocol. Students were paired randomly within their groups and given 45 minutes to complete the negotiation in dedicated recording studios. Each negotiation was video-recorded with multiple camera angles to capture verbal and non-verbal communication. Three independent experts with over five years of business negotiation experience evaluated all recordings using a validated 50-point rubric (five competencies: preparation, strategy, communication, adaptability, value creation; each scored 1–10 points), with these five dimensions confirmed by two internal university experts and one experienced industry practitioner with rich work experience. The experts underwent comprehensive training on the assessment criteria and achieved an inter-rater reliability coefficient (ICC) of 0.87 before formal scoring, and remained blinded to group assignment throughout the scoring process.

Week 1 featured the pre-test assessment, followed by 8 weeks of identical theoretical instruction for both groups (Weeks 2-9). This theoretical phase covered fundamental negotiation concepts during 24 class hours, addressing essential concepts including BATNA, ZOPA, value creation, distributive versus integrative strategies, and effective communication techniques. The consistent curriculum delivery across groups controlled for theoretical knowledge variation while isolating the practical intervention as the experimental variable.

The practical intervention phase occurred from Week 10 to Week 17, comprising 24 class hours of practice activities. The experimental group accessed the AI negotiation coach through a customized implementation on the Chaoxing learning platform, featuring dynamically adaptive opponents capable of assuming various negotiation styles and providing real-time, personalized feedback. Students completed multiple negotiation cases during this period, with the system logging all interactions for subsequent analysis. Meanwhile, the control group participated in traditional peer-to-peer role-playing activities, negotiating identical cases but with human partners while receiving standard instructor feedback following established pedagogical practices.

Week 18 featured the post-test assessment using the same case with switched roles. Performance was again video-recorded and evaluated by the same three independent experts using the identical rating scale across all six competencies, while maintaining blinding to group assignment.

Statistical analyses used SPSS 28, with descriptive statistics for all variables. Independent t-tests compared post-test scores between groups (RQ1), and paired t-tests examined pre-post differences within groups (RQ2 & RQ3).

III. FINDINGS

The descriptive statistics for both the control and experimental groups, each consisting of 45 participants, indicate notable changes from pre-test to post-test across the measured constructs. For the control group, pre-test means were 4.00 for Preparation, 4.96 for Strategy, 5.33 for Communication, 3.89 for Adaptability, 3.58 for Value creation, and 4.35 for Outcome, which increased to post-test means of 5.02, 5.31, 6.27, 5.73, 5.18, and 5.50, respectively. In the experimental group, pre-test means started at 4.18 for Preparation, 5.13 for Strategy, 5.38 for Communication, 3.80 for Adaptability, 3.69 for Value creation, and 4.44 for Outcome, and showed greater improvement in post-test means, reaching 6.67 for Preparation, 6.33 for Strategy, 5.02 for Communication, 5.44 for Adaptability, 6.62 for Value creation, and 6.02 for Outcome.

TABLE I. DESCRIPTIVE STATISTICS

Constructs	*Post-test*		*Pre-test*	
	Mean	*SD*	*Mean*	*SD*
Control Group				
Preparation	5.0222	.94120	4.0000	.87905
Strategy	5.3111	.92496	4.9556	.87790
Communication	6.2667	1.13618	5.3333	1.12815
Adaptability	5.7333	.78044	3.8889	.93474
Value creation	5.1778	.96032	3.5778	1.15776
Outcome	5.5022	.55861	4.3511	.88128
Experimental Group				
Preparation	6.6667	.92932	4.1778	1.00654
Strategy	6.3333	1.02247	5.1333	1.03573
Communication	5.0222	.83907	5.3778	1.36995
Adaptability	5.4444	1.30655	3.8000	.96766
Value creation	6.6222	1.07215	3.6889	1.20269
Outcome	6.0178	.73463	4.4356	.87312

Research question 1: The independent-samples t-test revealed significant differences in post-test performance between the experimental group using the AI negotiation coach and the control group using traditional methods across most competencies. Students in the experimental group demonstrated substantially higher scores in Preparation (mean difference = -1.64, p < .001), Strategy (mean difference = -1.02, p < .001), and Value Creation (mean difference = -1.44, p < .001), while the control group showed superior performance in Communication (mean difference = 1.24, p < .001). No statistically significant difference was found in Adaptability between the two groups (mean difference = 0.29, p = .207). For the comprehensive Outcome measure, which represents an average across all competencies, the experimental group achieved significantly higher results (mean difference = -0.52, p < .001), indicating an overall advantage for the AI-assisted training approach despite the control group's stronger performance in communication skills.

TABLE II. INDEPENDENT-SAMPLES T TEST

Constructs	*t*	*df*	*Two-Sided p*	*Mean Difference*	*Std. Error Difference*
Preparation	-8.340	88	<.001	-1.64444	.19717
Strategy	-4.973	88	<.001	-1.02222	.20554
Communication	5.910	80.991	<.001	1.24444	.21055
Adaptability	1.273	71.853	.207	.28889	.22687
Valuecreation	-6.732	88	<.001	-1.44444	.21457
Outcome	-3.747	82.133	<.001	-.51556	.13758

Research question 2: The paired-samples t-test for the experimental group revealed statistically significant improvements from pre-test to post-test across most competencies. Students demonstrated substantial gains in Preparation (mean difference = 2.49), Strategy (mean difference = 1.20), Adaptability (mean difference = 1.64), and Value Creation (mean difference = 2.93), with all showing significance at p < .001. No statistically significant change was observed in Communication skills (mean difference = -0.36, p = .156). The comprehensive Outcome measure, representing an average across all competencies, showed significant improvement (mean difference = 1.58, p < .001), indicating that despite the lack of progress in communication abilities, the AI coaching intervention produced substantial overall enhancement in negotiation performance for the experimental group.

TABLE III. PAIRED-SAMPLES T TEST IN EXPERIMENTAL GROUP

Experimental Group (Post-Pre)		*Mean*	*Std. Deviation*	*t*	*df*	*Two-Sided p*
Pair 1	Preparation	2.48889	1.34202	12.441	44	<.001
Pair 2	Strategy	1.20000	1.43970	5.591	44	<.001
Pair 3	Communication	-.35556	1.65359	-1.442	44	.156
Pair 4	Adaptability	1.64444	1.68085	6.563	44	<.001
Pair 5	Value creation	2.93333	1.52852	12.874	44	<.001
Pair 6	Outcome	1.58222	1.11667	9.505	44	<.001

Research question 3: The paired-samples t-test for the control group revealed statistically significant improvements from pre-test to post-test across most competencies, though with notably smaller effect sizes compared to the experimental group. Students demonstrated meaningful gains in Preparation (mean difference = 1.02, p < .001), Communication (mean difference

= 0.93, p < .001), Adaptability (mean difference = 1.84, p < .001), and Value Creation (mean difference = 1.60, p < .001). However, no statistically significant improvement was observed in Strategy (mean difference = 0.36, p = .062), indicating that traditional role-playing methods were insufficient for developing strategic negotiation capabilities. The comprehensive Outcome measure, representing an average across all competencies, showed significant improvement (mean difference = 1.15, p < .001), confirming that while traditional methods yielded overall progress, they were particularly ineffective in enhancing strategic thinking compared to other negotiation competencies.

TABLE IV. PAIRED-SAMPLES T TEST IN CONTROL GROUP

Control Group (Post-Pre)		*Mean*	*Std. Deviation*	*t*	*df*	*Two-Sided p*
Pair 1	Preparation	1.02222	1.13796	6.026	44	<.001
Pair 2	Strategy	.35556	1.24600	1.914	44	.062
Pair 3	Communication	.93333	1.48324	4.221	44	<.001
Pair 4	Adaptability	1.84444	1.24235	9.959	44	<.001
Pair 5	Value creation	1.60000	1.54331	6.955	44	<.001
Pair 6	Outcome	1.15111	1.00739	7.665	44	<.001

IV. DISCUSSION

The findings of this study provide robust empirical evidence supporting the efficacy of generative AI as a targeted pedagogical tool for developing specific negotiation competencies. The significantly greater improvement observed in the experimental group across preparation, strategy, and value creation demonstrates that AI negotiation coaches can effectively address well-documented gaps in traditional business education. These results align with prior research suggesting that AI-driven simulations provide opportunities for deliberate practice that are difficult to replicate in traditional classroom settings [5][9]. The AI system's ability to provide immediate feedback and adapt to student responses appears to have particularly enhanced strategic thinking and deal-design capabilities, which are essential for complex negotiation scenarios.

The differential outcomes across competencies offer important insights into the specific strengths and limitations of AI-based negotiation training. While the experimental group showed remarkable gains in cognitive and strategic domains (preparation, strategy, and value creation), their performance in communication skills did not improve significantly and actually lagged behind the control group. This pattern suggests that AI systems, while excellent for developing analytical and strategic competencies, may be less effective for cultivating interpersonal communication skills that require human interaction and nonverbal cues. This finding corroborates concerns raised by [9] regarding the limitations of AI in replicating the nuanced aspects of human communication.

Notably, the traditional role-playing approach used in the control group produced its strongest results in communication development, while showing minimal improvement in strategic thinking. This pattern highlights a fundamental limitation of conventional methods: while peer interactions effectively develop communicative abilities, they often lack the sophistication to challenge students' strategic assumptions or provide consistent, high-quality feedback on negotiation strategy. The absence of significant improvement in strategy for the control group (p = .062) underscores this critical gap in traditional pedagogy and emphasizes the unique value that AI coaches bring to strategic skill development.

The adaptability results, showing no significant difference between groups despite both demonstrating improvement, suggest that both approaches can develop this competency effectively. However, the mechanism likely differs—AI systems may enhance adaptability through exposure to diverse negotiation scenarios and styles, while human role-playing may develop it through unpredictable social interactions. This finding indicates that adaptability might be developed through multiple pathways, making it a particularly robust competency across different learning modalities.

This study has several limitations that should be acknowledged. The single-institution design within China may affect the generalizability of the results, suggesting a need for replication across diverse cultural and institutional contexts. Additionally, the relatively short duration of the intervention (one semester) warrants longer-term investigations to assess skill retention and real-world transfer ability.

While the AI system effectively enhanced strategic and cognitive skills, its limited impact on communication competencies highlights the need for future research on hybrid pedagogical models. Such approaches could integrate AI-driven strategic training with human-facilitated communication practice to leverage the strengths of both methods. Further studies should also explore the specific mechanisms behind AI coaching efficacy—such as feedback timing and adaptive algorithms—and examine how individual differences influence learning outcomes to enable more personalized applications.

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